

Detector Ranking is Not Architecture-Invariant under Medical Covariate Shift

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Abstract

Post-hoc out-of-distribution (OOD) detectors are almost always ranked on a single backbone. We ask how much that ranking moves when only the backbone changes, holding one medical covariate split fixed. On Camelyon17 hospital-2 (WILDS), Kendall’s $W = 0.694$ across eight CNN families and seven scores (seed 42); MSP AUROC spans 0.515 (ResNet-50) to 0.883 (DenseNet-121), and the DenseNet MSP jump holds on 5/5 seeds. But retraining one fixed architecture with a different seed reorders the seven scores almost as much: cross-seed W is 0.600–0.931 (mean 0.780), so the full-ranking reorder cannot be attributed to architecture alone. **What is stable is the feature-space family:** Mahalanobis or k NN is rank 1 on every backbone and every seed tested. Skin ISIC→PAD does not reproduce the MSP jump ($W = 0.791$). AUROC is not a proxy for deployment coverage: the sign of the AUROC versus coverage@risk 10% disagreement flips between Camelyon and MIDOg, and on a patient-disjoint split of hospital 2, choosing the backbone by validation coverage@risk instead of validation AUROC raises held-out coverage from 0.539 to 0.754. A pre-registered ViT-B/16 MSP of 0.884 met a locked threshold of 0.6947. The protocol: default to feature-space scores, do not copy a logit-score ranking across backbones or seeds, and pick a triage backbone from coverage@risk on a target-domain validation split, not from an AUROC table.

Keywords: out-of-distribution detection, domain shift, histopathology, dermoscopy, benchmark evaluation, model selection

1. Introduction

We test one hypothesis: *under a fixed medical covariate shift, post-hoc detector ranking is an architecture property*. On Camelyon17 hospital-2, ResNet-50 MSP AUROC is 0.515 while DenseNet-121 is 0.883 (Figure 1a). A lab that copies an OpenMIBOOD-style ResNet-50 recipe and a lab that uses DenseNet are not ranking the same detectors. And AUROC order does not predict coverage@risk order, so an AUROC table is not a safe guide to triage: on a patient-disjoint split, selecting by AUROC instead of coverage@risk loses 0.215 held-out coverage at a 10% risk budget (Figure 1b shows the retrospective full-test-set version). The answer we reach is narrower than the hypothesis: retraining one architecture with a new seed reorders the scores about as much as changing the architecture, and only the feature-space scores are stable across both.

That hypothesis is *not* that ranking was previously assumed architecture-invariant. On natural-image semantic OOD, ranking already moves with architecture (Zhang et al., 2023; Szyk et al., 2023; Datko et al., 2026; Claros Olivares and Brockmeier, 2025). OpenMIBOOD (Gutbrod et al., 2025) already showed that ImageNet method ranking does not

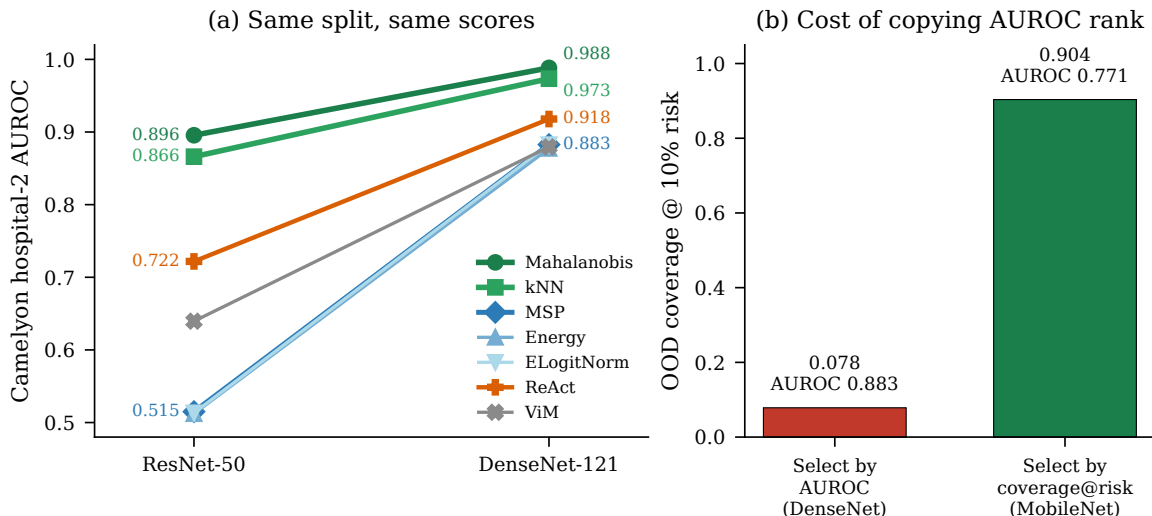


Figure 1: **The claim in one figure.** (a) Same Camelyon hospital-2 split, same seven scores: switching ResNet-50→DenseNet-121 lifts logit AUROC by ~ 0.37 while Mahalanobis and k NN stay on top. (b) On the full labeled test set (retrospective), the AUROC-highest backbone (DenseNet) has coverage@risk 10% = 0.078 while the coverage@risk-highest backbone (MobileNet) reaches 0.904: AUROC rank and coverage rank disagree. Section 4.5 gives the held-out version.

transfer to medical data and that feature-space scores outperform logit scores under a frozen classifier per domain. The orthogonal question is: **holding one medical covariate-shift dataset fixed, how much does ranking move when only the backbone changes?**

The rest of the paper is that test, in order. Camelyon discovers the shift: Kendall $W = 0.694$ and a backbone-dependent MSP jump. Five seeds verify the DenseNet MSP jump (5/5), and a cross-seed W bounds how much of the ranking reorder is training noise. Matched-recipe training asks whether that jump is architecture or optimizer. Skin falsifies a universal MSP-jump law, while feature ranks stay 1–2. MIDOG is the deployment readout: the *sign* of AUROC versus coverage@risk disagreement flips. A pre-registered ViT-B/16 tests whether the Camelyon MSP jump extends past CNNs. The contribution is a protocol, not a detector.

Contributions.

1. **On Camelyon, logit scores reorder across backbones—and across seeds.** Official Kendall $W = 0.694$ ($n=8$, 7 methods, seed 42). MSP AUROC span 0.515–0.883 matches the Maha–MSP gap on ResNet-50. Copying ResNet-50’s winning logit (ReAct) onto ConvNeXt costs 0.185 AUROC. The seed-robust MSP jump is DenseNet-121 (0.265 ± 0.065 , 5/5); ConvNeXt and EffV2-S@224 are 4/5, MobileNet/RegNet 3/5, EffB3 2/5. Cross-seed W at fixed architecture is 0.600–0.931, so the full-ranking re-

Table 1: Closest ranking-vs-backbone papers. Architecture-sensitive ranking exists on natural-image semantic OOD; the increment here is a held-fixed medical covariate split, zoo-scale W , and operational coverage.

Work	#arch.	Shift	Rank stat.	Utility
OpenOOD v1 (Yang et al., 2022)	1/set	semantic	—	AUROC
OpenOOD v1.5 (Zhang et al., 2023)	3	semantic	qualitative	AUROC
Szyc et al. (Szyc et al., 2023)	3 var.	CIFAR sem.	rank Δ	AUROC
Datko et al. (Datko et al., 2026)	per line	natural	claimed stable	AUROC
OpenMIBOOD (Gutbrod et al., 2025)	1/domain	medical	—	AUROC
Ours	8+ViT	covar.	Kendall W	AUROC+cov.

order is not architecture alone; feature-space scores hold rank 1 in all 20 anchor×seed cells.

2. **Skin falsifies universality.** ISIC→PAD ($n=8$, $W=0.791$): Maha rank-1 and k NN rank-2 on all eight; MSP stays in a narrow band. Not a second Camelyon jump.
3. **AUROC is not a reliable proxy for deployment coverage.** The sign of the AUROC versus coverage@risk 10% disagreement flips between Camelyon and MIDOG (Geifman and El-Yaniv, 2019; Traub et al., 2024). On a patient-disjoint split of Camelyon hospital 2, selecting by validation coverage@risk instead of validation AUROC raises held-out coverage from 0.539 to 0.754 (Section 4.5); the larger full-test-set gap (0.826) is retrospective only.
4. **A locked prediction.** Before training: ViT-B/16 Camelyon MSP ≥ 0.6947 . Observed 0.884. Held out of the 8-CNN W .

2. Related work

OpenMIBOOD (Gutbrod et al., 2025) ranks 24 post-hoc detectors with a **single frozen classifier per medical domain**. It shows that ImageNet method ranking does not transfer to medical data and that feature-space scores outperform logit scores under that protocol. Table 1 locates the increment: we hold the dataset fixed and change the backbone.

OpenOOD v1 (Yang et al., 2022) fixes architecture on purpose. OpenOOD v1.5 (Zhang et al., 2023) compares three ImageNet models on semantic near/far OOD and reports that “different post-processor may favor different architecture,” without Kendall’s W and without a covariate hospital split. Szyc et al. (Szyc et al., 2023) show rankings change across three ResNet CIFAR implementation variants. Datko et al. (Datko et al., 2026) claim an architecture-specific winner that transfers across natural-image OOD—the opposite of a single-split concordance statistic.¹ Claros Olivares and Brockmeier (Claros Olivares and Brockmeier, 2025) rank confidence scores with AURC/AUGRC on CIFAR / TinyImageNet;

1. This citation could not be independently verified against a primary source at the time of writing; see `refs.bib`. Confirm the original DOI/URL before submission.

they do not report a sign-flip of AUROC versus coverage@risk 10% across medical covariate domains.

OoD-Bench (Ye et al., 2022) is OOD generalization, not detection. WILDS (Koh et al., 2021) supplies Camelyon17 and iWildCam; iWildCam is a negative bound, not a third main domain.

3. Setup

3.1. Shifts

Camelyon17 (Bándi et al., 2019), via the WILDS (Koh et al., 2021) split: hospitals 0, 3, 4 as ID and hospital 2 as OOD (WILDS’s official OOD-test domain); founding cell, official W , $n=8$. Hospital 1 is WILDS’s official OOD-*validation* domain; we do not use it for any number in the main paper and report it only in the held-out stain-jitter check (Appendix A). Skin ISIC→PAD (Codella et al., 2019; Pacheco et al., 2020): feature rank-stability without an MSP jump. MIDOG (Aubreville et al., 2023): OpenMIBOOD’s (Gutbrod et al., 2025) 1a vs. 1b+1c split and checkpoints, coverage sign-flip; official W is $n=3$. We build on OpenMIBOOD’s public MIDOG split and ResNet-50 checkpoint as an external reference. Public R50 AUROC ~ 0.59 is *theirs*; in-house R50 MSP 0.512 is *ours*. We never mix the two.

3.2. Backbones

Eight CNN-family models, ImageNet initialization, 10-epoch fine-tune, seed 42, train-frac 1.0 on Camelyon: ResNet-18/50 (He et al., 2016), DenseNet-121 (Huang et al., 2017), ConvNeXt-Tiny (Liu et al., 2022), MobileNetV3-Large (Howard et al., 2019), RegNetY-3.2GF (Radosavovic et al., 2020), EfficientNet-B3 (Tan and Le, 2019) at native 300, and EfficientNetV2-S (Tan and Le, 2021) at **224** (native 384 dropped). Standard fine-tune recipes (Adam / AdamW / SGD) follow each family’s common practice and are *not* forced-identical Adam. That confound is measured in Section 4.3. Only resolution was isolated in a controlled pair (EffB3 @224 vs. @300; EffV2-S @224 vs. @384). ViT-B/16 (Dosovitskiy et al., 2021) @224 is held out of W .

3.3. Scores and statistics

Seven post-hoc scores: MSP (Hendrycks and Gimpel, 2017), Energy (Liu et al., 2020), energy of logit-normed logits (ELogitNorm), ViM (Wang et al., 2022), ReAct (Sun et al., 2021), Mahalanobis with Ledoit–Wolf covariance (Lee et al., 2018; Ledoit and Wolf, 2004), and k NN (Sun et al., 2022). Primary concordance: **Kendall’s W per domain, never pooled**. Jump: $\text{MSP} - \text{mean}(\text{R18}, \text{R50}) \geq 0.15$. Feature drag: Maha or k NN on a new backbone below $\text{median}(\text{R18}, \text{R50}, \text{EffB3}) - 0.10$. Official Camelyon ResNet mean MSP = 0.544654; jump iff $\text{MSP} \geq 0.695$ on that split.

Coverage@risk 10% is OOD-triage: rank OOD samples by MSP (higher = more ID-like) and take the largest coverage whose classifier risk is $\leq 10\%$. ID-selective Camelyon is vacuous (ID-val acc $\approx 99.5\%$) and is not cited. AUROC and coverage@risk measure different things: AUROC is ID-vs-OD separability of the detector score alone, while coverage@risk 10% depends jointly on that ranking *and* the classifier’s own accuracy on the OOD domain, so

Table 2: Camelyon hospital-2 AUROC, **seed 42** (official W). Jump iff $\text{MSP} \geq 0.6947$ vs. frozen ResNet mean 0.544654. MSP range is ResNet-50 $\rightarrow$ DenseNet.

Backbone	input	Maha (rank)	k NN (rank)	MSP	seed-42 jump
ResNet-18	224	0.956 (1)	0.936 (2)	0.574	no
ResNet-50	224	0.896 (1)	0.866 (2)	0.515	no
DenseNet-121	224	0.988 (1)	0.973 (2)	0.883	yes
ConvNeXt-Tiny	224	0.928 (1)	0.916 (2)	0.846	yes
MobileNetV3-L	224	0.870 (2)	0.904 (1)	0.771	yes
RegNetY-3.2GF	224	0.964 (1)	0.956 (2)	0.784	yes
EfficientNet-B3	300	0.885 (2)	0.934 (1)	0.803	yes
EfficientNetV2-S	224	0.880 (1)	0.856 (2)	0.778	yes

the two need not agree, and a disagreement between them is not itself surprising without separately checking which one moved. A comparison of the two on the full labeled OOD test set is retrospective; Section 4.5 therefore also selects on a patient-disjoint validation half and evaluates on the other half.

We tested whether the Camelyon MSP pattern is an artifact of input resolution, temperature scaling (Guo et al., 2017), ID-only detector *selection*, a CE+SupCon recipe (Khosla et al., 2020), or train-time HED stain jitter. None is a clean account (Section 5 and Appendix A).

4. Results

4.1. Camelyon-17 discovers the ranking shift

Official Kendall $W = \mathbf{0.694}$ ($n=8$, 7 methods). This is the only Camelyon number called “official W .” Leave-one-backbone recomputation of W ranges 0.676–0.738; that is *structural* sensitivity of the concordance statistic, not a 95% CI (Appendix A).

Feature methods occupy ranks 1–2 on every official backbone (Maha #1 on 6/8; k NN #1 on EffB3 and MobileNet). No backbone drags Maha/ k NN past the 0.10 feature-drag threshold. On Table 2, MSP spans 0.515–0.883; Maha 0.870–0.988; k NN 0.856–0.973. Incomplete concordance ($W=0.694$) is the logit ranks moving (Figure 2).

Table 3 reports the full five-seed MSP matrix: seeds verify the DenseNet MSP jump. DenseNet-121 is the confirmed seed-robust jump (5/5). ConvNeXt-Tiny and EfficientNetV2-S@224 are not seed-42 flukes (4/5) and are not promoted to 5/5. MobileNetV3 and RegNetY are 3/5; EfficientNet-B3 is 2/5 (its seed-42 jump +0.258 is a favorable draw relative to 0.142 ± 0.125). The seed-42 matrix still shows a jump without feature drag. The claim is narrower than “6/6 non-ResNet, confirmed.”

Copying a logit ranking is not free. ResNet-50’s best logit is ReAct (0.722); on ConvNeXt the same score is 0.692 against Energy 0.877—a 0.185 AUROC transfer cost. Feature-space transfer regret is ≤ 0.05 because Maha/ k NN stay in ranks 1–2.

Table 3: Same eight Camelyon hospital-2 MSP cells on all five seeds {42, 43, 44, 45, 46}. Jump = MSP minus that seed’s ResNet mean; bar 0.15. The last column is how many of those five seeds clear the bar. Official W is seed 42 only.

Backbone	42	43	44	45	46	jump mean $\pm$ sd	≥ 0.15
ResNet-18	0.574	0.712	0.601	0.524	0.581	—	—
ResNet-50	0.515	0.621	0.619	0.515	0.661	—	—
ResNet mean	0.545	0.667	0.610	0.520	0.621	—	—
DenseNet-121	0.883	0.865	0.868	0.845	0.826	0.265 $\pm$ 0.065	5/5
ConvNeXt-Tiny	0.846	0.799	0.803	0.764	0.785	0.207 $\pm$ 0.067	4/5
MobileNetV3-L	0.771	0.728	0.815	0.780	0.745	0.175 $\pm$ 0.081	3/5
RegNetY-3.2GF	0.784	0.778	0.836	0.830	0.724	0.198 $\pm$ 0.089	3/5
EfficientNet-B3	0.803	0.731	0.643	0.818	0.678	0.142 $\pm$ 0.125	2/5
EfficientNetV2-S@224	0.778	0.744	0.808	0.846	0.835	0.210 $\pm$ 0.089	4/5

Cross-seed control. Is the incomplete concordance an architecture effect or training noise? For the four anchors trained on all five seeds we compute Kendall’s W over the seven scores with the seeds, not the backbones, as raters. Cross-seed W is 0.931 (ResNet-50), 0.800 (ResNet-18), 0.789 (DenseNet-121) and 0.600 (EfficientNet-B3); mean 0.780 ± 0.136 . EfficientNet-B3’s value is below the cross-architecture 0.694, and cross-architecture W over the same four anchors at a fixed seed is 0.750–0.915. Retraining one architecture therefore reorders the seven scores about as much as changing the architecture does, and the cross-architecture W cannot be read as an architecture effect on its own. What survives the control is narrower: a feature-space score is rank 1 in all 20 anchor $\times$ seed cells (both in ranks 1–2 in 18/20), and the DenseNet MSP jump (5/5 seeds) is an architecture effect on the MSP *level*, not on the full ranking. Backbone choice moves the logit leaderboard, but so does the seed; neither unseats the feature default.

4.2. Skin falsifies a universal MSP jump

Official $W = 0.791$ ($n=8$). Maha is rank-1 and k NN rank-2 on **all eight** (Figure 2). MSP sits in a narrow band (0.68–0.74). There is no Camelyon-style ResNet-vs-rest MSP jump. EffV2-S@224 MSP is 0.737; the native-384 cell that put k NN at rank 7 (MSP 0.823) is a resolution artifact and is not in official W . Skin is rank-stability of feature scores under a zoo: a replication of OpenMIBOOD’s feature $>$ logit pattern *across architectures*, not an MSP-jump.

4.3. Optimizer challenges a clean architecture reading

Resolution is not the Camelyon DenseNet / seed-42 pattern. EffB3 Camelyon: 0.803 @300 $\rightarrow$ 0.787 @224 (the jump versus the official ResNet mean survives). EffV2-S Camelyon: 0.726 @384 $\rightarrow$ **0.778** @224 (the official cell; the jump is not a 384 artifact). On skin, the EffV2-S @384 k NN collapse *is* a resolution artifact and disappears at 224. These two facts are not a slogan that “resolution never matters.”

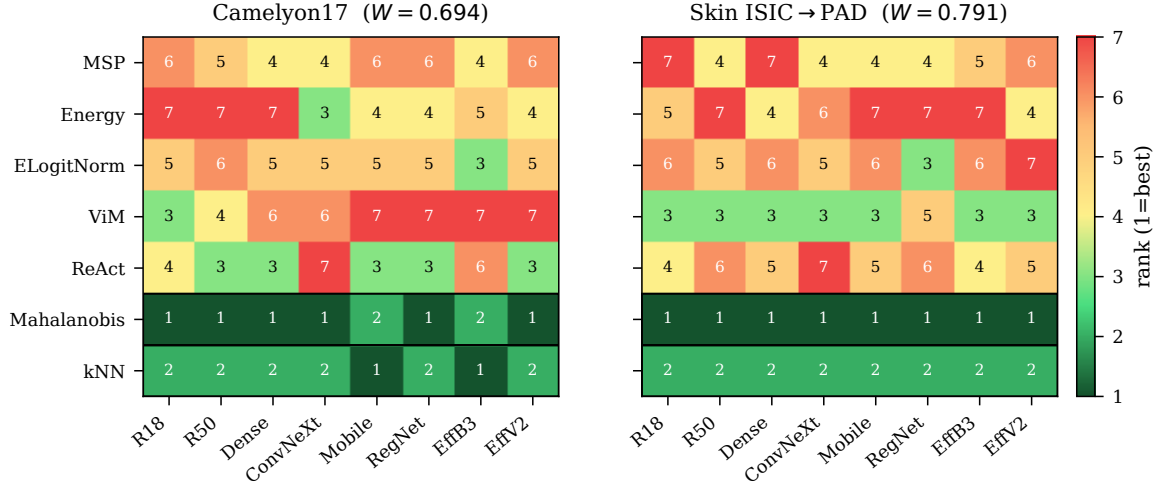


Figure 2: **Method rank by backbone** (1=best AUROC). Feature-space scores occupy ranks 1–2 on every official backbone in both domains. Logit ranks move on Camelyon ($W=0.694$) and much less on skin ($W=0.791$), where Maha is rank-1 on 8/8.

M1 — jumpers trained with ResNet Adam ($\text{lr}=10^{-4}$, $\text{wd}=10^{-4}$, batch 64). The four official zoo members whose standard recipe was not that Adam still jump versus the frozen official ResNet mean 0.545: ConvNeXt 0.820 (+0.275), MobileNetV3 0.737 (+0.193), RegNet 0.805 (+0.261), EffV2-S@224 0.873 (+0.328); all with id-val acc ≥ 0.994 . 4/4 clean hits. DenseNet and EffB3 already shared ResNet-Adam in the official zoo. Recipe mismatch does *not* explain the seed-42 jumps of those four members. M1 is seed 42 only; the five-seed MSP matrix is Table 3.

M2 — ResNets trained with ConvNeXt AdamW ($\text{wd}=0.05$). ResNet-18 MSP **0.773** (enters the 0.695 band; id-val 0.9957). ResNet-50 MSP 0.581 (stays out; id-val 0.9958). ResNet *can* jump under a non-default recipe. Architecture is not a clean variable, independent of optimizer. These two results point in opposite directions; we report both.

4.4. ViT-B/16: a locked prediction

Locked *before* training, from the seed-42 zoo rule that non-ResNet-like cells jumped: MSP ≥ 0.6947 . Linear depth/parameter/receptive-field predictors are invalid on this $n=8$ and are not quoted. Observed MSP **0.884**, Maha 0.958, k NN 0.941. The prediction holds. Not a ninth W column.

4.5. Coverage@risk: the deployment cost

Protocol: OOD-triage coverage@risk 10% with MSP, not ID-selective coverage.

On Camelyon hospital-2, the locked $n=4$ utility table has MSP AUROC order EffB3@300 (0.803) > EffB3@224 (0.787) > ResNet-18 (0.574) > ResNet-50 (0.515), but coverage@risk 10%

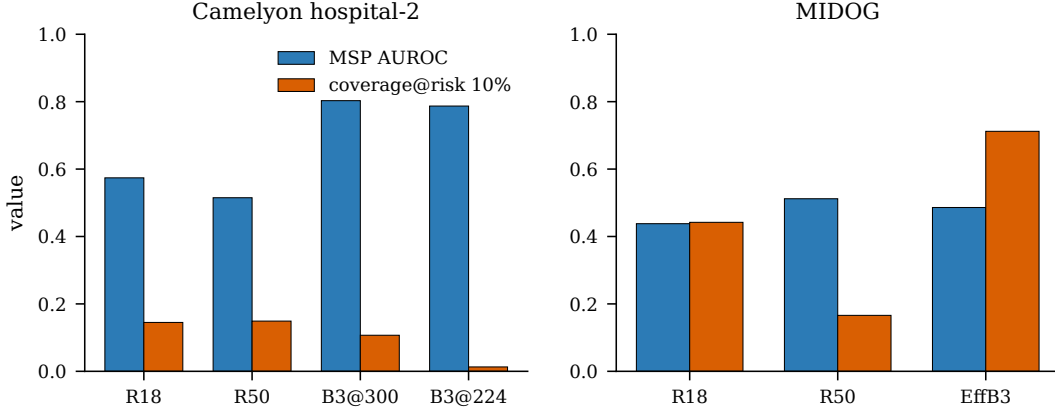


Figure 3: **AUROC vs. coverage@risk 10%**. On Camelyon the AUROC-high Efficient-Nets have the *lowest* triage coverage; on MIDOG the AUROC-high ResNet-50 is coverage-lowest. The sign of the disagreement flips.

order ResNet-50 (0.149) > ResNet-18 (0.145) > EffB3@300 (0.107) > EffB3@224 (0.013). The AUROC winner is not the coverage winner (Figure 3).

On MIDOG, official $n=3$ (in-house): MSP AUROC ResNet-50 0.512, EffB3 0.486, ResNet-18 0.438 (gaps are small). Coverage@risk 10%: EffB3 0.712, ResNet-18 0.442, ResNet-50 0.166. ResNet-50 is AUROC-highest and coverage-lowest. The permutation is *not* the Camelyon permutation.

AUROC/AUGRC already assume that AUROC and coverage-based ranking can disagree (Geifman and El-Yaniv, 2019; Traub et al., 2024). The increment is the *sign* of that disagreement flipping between two medical covariate domains under the same protocol, with no rule for which backbone wins. On the full labeled Camelyon test set, AUROC order and coverage@risk order on the $n=8$ zoo diverge by 0.826 coverage (DenseNet 0.078 vs. MobileNet 0.904; patch-bootstrap 95% CI of the difference [0.817, 0.837]; Figure 1b). That comparison is retrospective: it reads both rankings off the test labels, which do not exist at deployment time.

Held-out selection. To make it a procedure, we split the nine hospital-2 patients into two halves (stratified on the tumor label, grouped by patient, seed 42, frozen before any model was scored): six patients (41,217 patches) for selection, three (43,837) for evaluation. Selecting the backbone with the highest validation MSP AUROC picks ConvNeXt (0.877), which reaches coverage@risk 10% of 0.539 on the test half. Selecting by validation coverage@risk picks EfficientNetV2-S@224 (0.821), which reaches 0.754: a +0.215 gain. Both are well below the best test-half backbone (MobileNet, 1.000), and coverage varies strongly with which patients fall in each half (one patient contributes 31,878 patches, 85% tumor). Coverage-based selection on a target validation split helps, but at $n=8$ backbones and nine patients it is noisy, not solved.

4.6. Frozen foundation models

Camelyon linear-probe MSP does not invert: GigaPath (Xu et al., 2024) 0.664, Phikon (Filiot et al., 2023) 0.605, UNI (Chen et al., 2024) 0.599, all >0.5 . MIDOG UNI / GigaPath MSP 0.445 / 0.470 is a MIDOG-only footnote. FM Maha ~ 0.995 –1.0 is discussion support that feature $>$ logit can survive a training-paradigm change, not a pillar.

5. Failed alternatives

Each of the following was a precommitted test. None is hidden.

Resolution (from Section 4.3). EffB3@224 MSP 0.787; EffV2-S@224 MSP 0.778. Not a pixel-size explanation of Table 2.

Temperature T^* . Correlation of T^* with the MSP jump: $r=0.268$, $p=0.52$. DenseNet–ResNet-18 MSP gap 0.308 $\rightarrow$ 0.308 after T^* . Calibration/temperature is not the mechanism.

ID-only Maha/MSP gate. Selecting a score from ID-only statistics is a *selector*, not a detector. On Camelyon it picks Maha on 2/8 backbones; mean AUROC of the gated score 0.837 versus always-Maha 0.940. The bar was matching Maha and ViM. It does not.

CE+SupCon ($\lambda=0.1$, $\tau=0.07$), ResNet-18 + DenseNet. Gap 0.308 $\rightarrow$ 0.238 (23%; pre-commit bar was 50% / ≤ 0.154). ResNet-18 MSP 0.574 $\rightarrow$ **0.515** (worse; this is the SupCon cell, not official ResNet-50 MSP 0.515). Part of the shrink is the low model degrading. Dirty; not a recipe.

HED stain jitter, an in-house MIDOG $n=8$ zoo, and a shift-type extract (covariate / task / far-OOD) are in Appendix A. Joint readout: no shift-type law, no stain mitigation, no Camelyon-style MSP jump on MIDOG.

6. Discussion and limitations

The hypothesis holds only in a narrow form, and fails as a cross-medical MSP-jump law. On Camelyon, backbone choice changes the logit-score ranking and produces a seed-robust DenseNet MSP jump, but retraining one architecture reorders the seven scores about as much, so a cross-architecture Kendall’s W (0.694) is not by itself evidence of an architecture effect. The stable finding is the feature-space family: Mahalanobis or k NN is rank 1 on every backbone and every seed we tested, when the shift looks like Camelyon *and* when it does not (skin). M1 preserves the seed-42 jumps under ResNet-Adam; M2 lets ResNet-18 enter the MSP band under ConvNeXt-AdamW: architecture is the dominant axis, not a clean one. ViT-B/16 was a locked prediction, not a post-hoc ninth CNN.

The practical cost is not abstract. Copying ResNet-50’s logit winner onto ConvNeXt costs 0.185 AUROC, and the cross-seed result implies a logit ranking does not reliably transfer even between seeds of one backbone. Selecting a triage backbone from AUROC rather than from coverage@risk on a target validation split loses 0.215 held-out coverage at a 10% risk budget. For benchmarks, this argues for reporting rank concordance across seeds as well as across backbones before reading a detector leaderboard, and for reporting coverage@risk on a held-out target split next to AUROC.

Limitations. (1) **iWildCam bound**. Locked 3-backbone cell: DenseNet MSP 0.575 vs. ResNet mean 0.604, jump -0.029 . Extra five CNNs vs. that domain’s ResNet mean: Con-

vNeXt 0.601, MobileNet 0.558, RegNet 0.580, EffB3 0.577, EffV2-S 0.650; 0/5 jump. The Camelyon MSP-jump is absent on this non-medical covariate shift and may be biomedical-specific. iWildCam is not pooled with Camelyon W . (2) **No clean mitigation.** CE+SupCon is dirty (ResNet-18 MSP worse). HED stain jitter is dirty (DenseNet MSP 0.780<0.80). The ID-only gate is a selector. (3) **MIDOG official W is $n=3$ ($W=0.857$);** the in-house $n=8$ zoo shows no MSP jump and does not replace it. (4) W is per-domain; 0.694 / 0.791 / 0.857 are not averaged. (5) No reader study: the operational claim stops at coverage@risk. Patch-bootstrap 95% CIs for Camelyon coverage are narrow (DenseNet 0.078 [0.067, 0.086], MobileNet 0.904 [0.898, 0.909]) but ignore patient clustering; MIDOG ResNet-50 coverage 0.166 [0.004, 0.369] is wide. (6) Architecture is confounded with its standard recipe. 5/5 seed-robustness is DenseNet only. Official W stays seed 42. Staining vs. scanner vs. tissue prep inside the hospital shift is not isolated. (7) **Held-out selection is one split of nine patients.** The +0.215 gain comes from a single frozen patient split; repeated patient-level resampling was not run, and MIDOG ($n=3$) is not split. (8) **Seed variance is comparable to architecture variance in the full ranking.** Cross-seed W (0.600–0.931) covers four anchors on Camelyon only; it was not computed for skin or MIDOG. Any claim about the full 7-score ranking, rather than about the feature-family default or the MSP level, is confounded with training seed.

Code for the architecture-zoo evaluation and coverage@risk will be released, crediting OpenMIBOOD for MIDOG imglists, the public ResNet-50 checkpoint, and the 1a / cs-ID / near-OOD protocol.

7. Conclusion

Under medical covariate shift, logit-based detector scores reorder with the backbone, but about as much with the training seed; the feature-space family stays on top across both. On Camelyon that is a seed-robust DenseNet MSP jump, stable feature ranks, and a 0.215 held-out coverage gain from selecting a backbone by validation coverage@risk instead of AUROC. Skin keeps the feature ranks and drops the MSP jump. MIDOG flips the sign of the AUROC–coverage disagreement. A pre-registered ViT hit shows the Camelyon MSP jump is not a CNN-zoo accident; iWildCam does not reproduce it. The protocol is the result: default to feature-space scores, do not copy a logit-score ranking across backbones or seeds, and estimate a backbone’s coverage@risk on a held-out validation split of the deployment shift, not from an AUROC table.

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Appendix A. Supplementary material

Leave-one-backbone W is not a CI. Dropping one backbone and recomputing Cameleon W on the remaining seven yields 0.676–0.738 (largest increase when dropping ConvNeXt; largest decrease when dropping EffV2-S or RegNet). This is structural sensitivity of the concordance statistic, not a seed confidence interval; the seed question is answered by the cross-seed control in Section 4.1.

HED stain jitter. Train-only HED stain jitter ($\sigma=0.20$, $p=1.0$) on ResNet-18, ResNet-50 and DenseNet-121, same Adam recipe as official, seed 42. Precommit dirty-guard: DenseNet hospital-2 MSP $<0.80 \Rightarrow$ dirty. Hospital-2 MSP: ResNet-18 0.772, ResNet-50 0.593, DenseNet 0.780 (below 0.80); gap DenseNet–ResNet-18 0.008; id-val acc ≥ 0.994 . Held-out hospital-1 MSP: ResNet-18 0.765, ResNet-50 0.685, DenseNet 0.796. Dirty: part of the gap shrink is DenseNet falling (0.883 $\rightarrow$ 0.780) while ResNet-18 rises. Not a mitigation.

MIDOG in-house zoo $n=8$ (not official W). Same 1a vs. 1b+1c protocol, not mixed with public OpenMIBOOD R50 ~ 0.59 . MSP / Maha: ResNet-18 0.438 / 0.733, ResNet-50 0.512 / 0.839, DenseNet-121 0.447 / 0.910, ConvNeXt-Tiny 0.521 / 0.801, MobileNetV3-L 0.469 / 0.714, RegNetY-3.2GF 0.553 / 0.944, EfficientNet-B3 0.486 / 0.673, EfficientNetV2-S@224 0.493 / 0.581. MSP 0.44–0.55 on all eight; 0/6 non-ResNet jumps. A domain-scope bound, not a replacement of official W ($n=3$, 0.857).

Shift-type extract. Same eight Camelyon seed-42 classifiers, ID = Camelyon id_val. Arm C (hospital-2, covariate): 6/6 non-ResNet jumps. Arm T (MIDOG 1a, task shift): 0/6. Arm F (CIFAR-10 test, far-OOD): 2/6. Per-arm Kendall W (never pooled, not official): C 0.731, T 0.692, F 0.846. No shift-type law.

Five-seed jumps. Figure 4 plots each backbone’s MSP minus that seed’s ResNet mean; the underlying MSP matrix is Table 3.

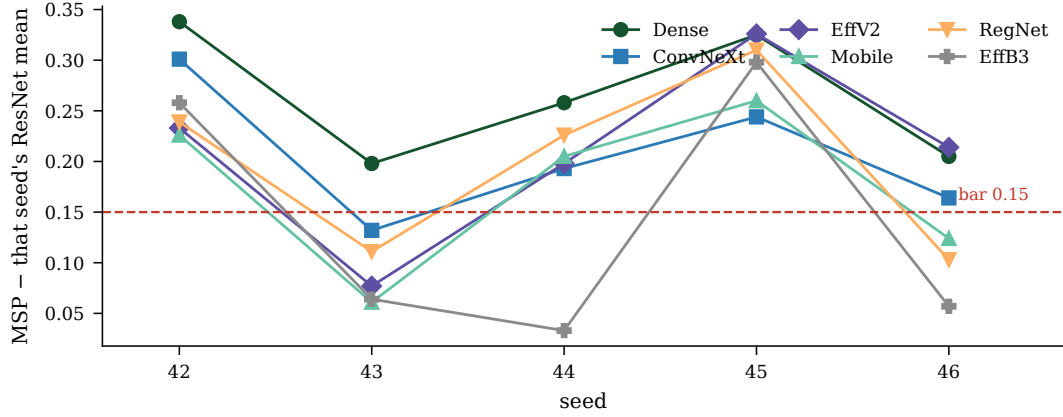


Figure 4: Per-seed MSP jump versus that seed’s ResNet mean. DenseNet stays above 0.15 on all five seeds; EfficientNet-B3 does not.

Other closed checks. ECE, BN-drift, eigenspace summaries, n_{train} sweeps and CIFAR-10-C were closed earlier and are not pillars.